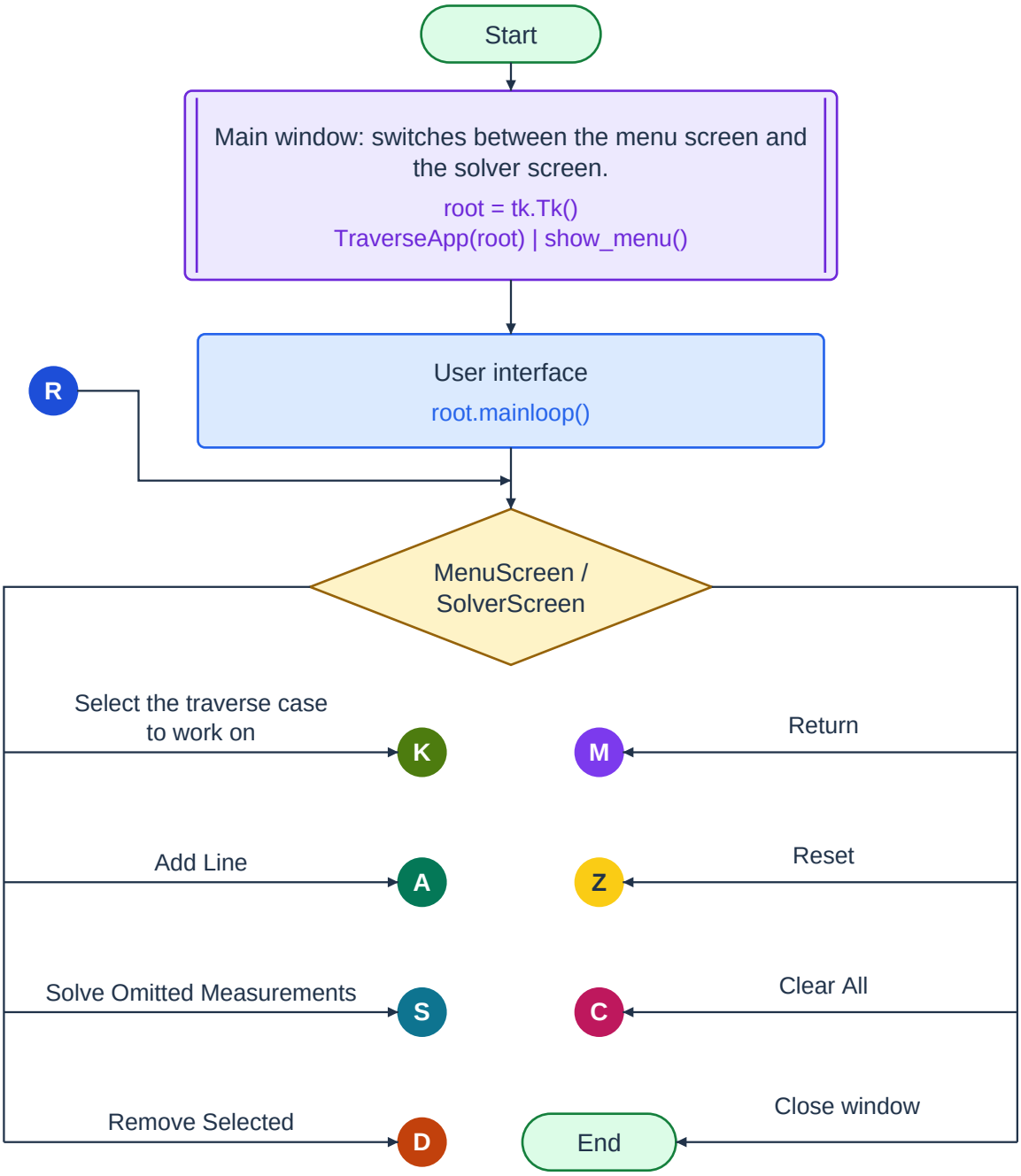
1 User interface



NUMBERED STEPS

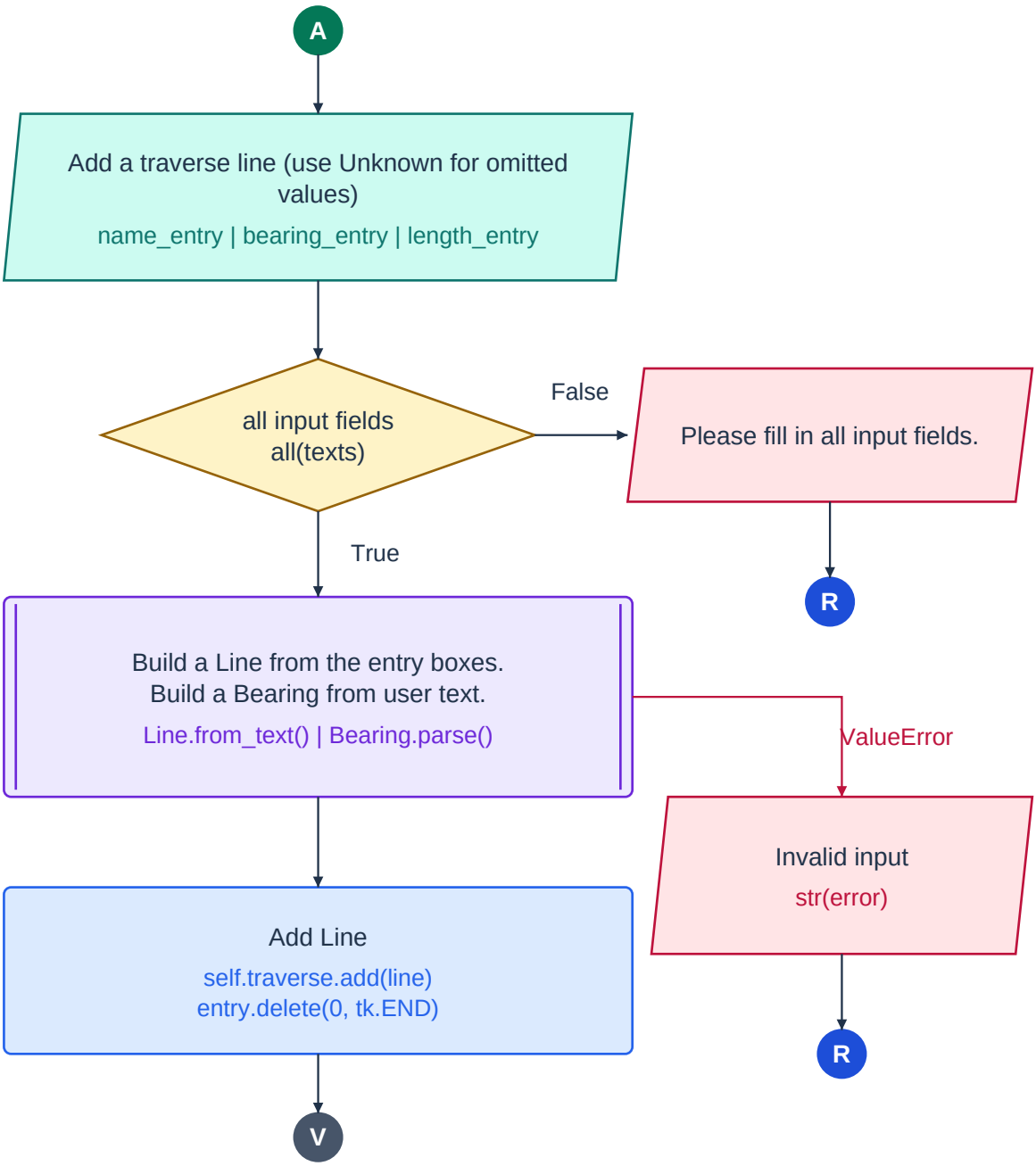
1. Start. Create root = tk.Tk(), then TraverseApp(root); show_menu() displays the menu.

2. User interface. Run root.mainloop() before any case-selection or button callback.

3. Select the traverse case to work on. K opens a case; A, S, D, Z and C perform the selected action.

4. Return / End. M shows the menu. R resumes the event loop. Close window is the only path to End.

2 Add Line



NUMBERED STEPS

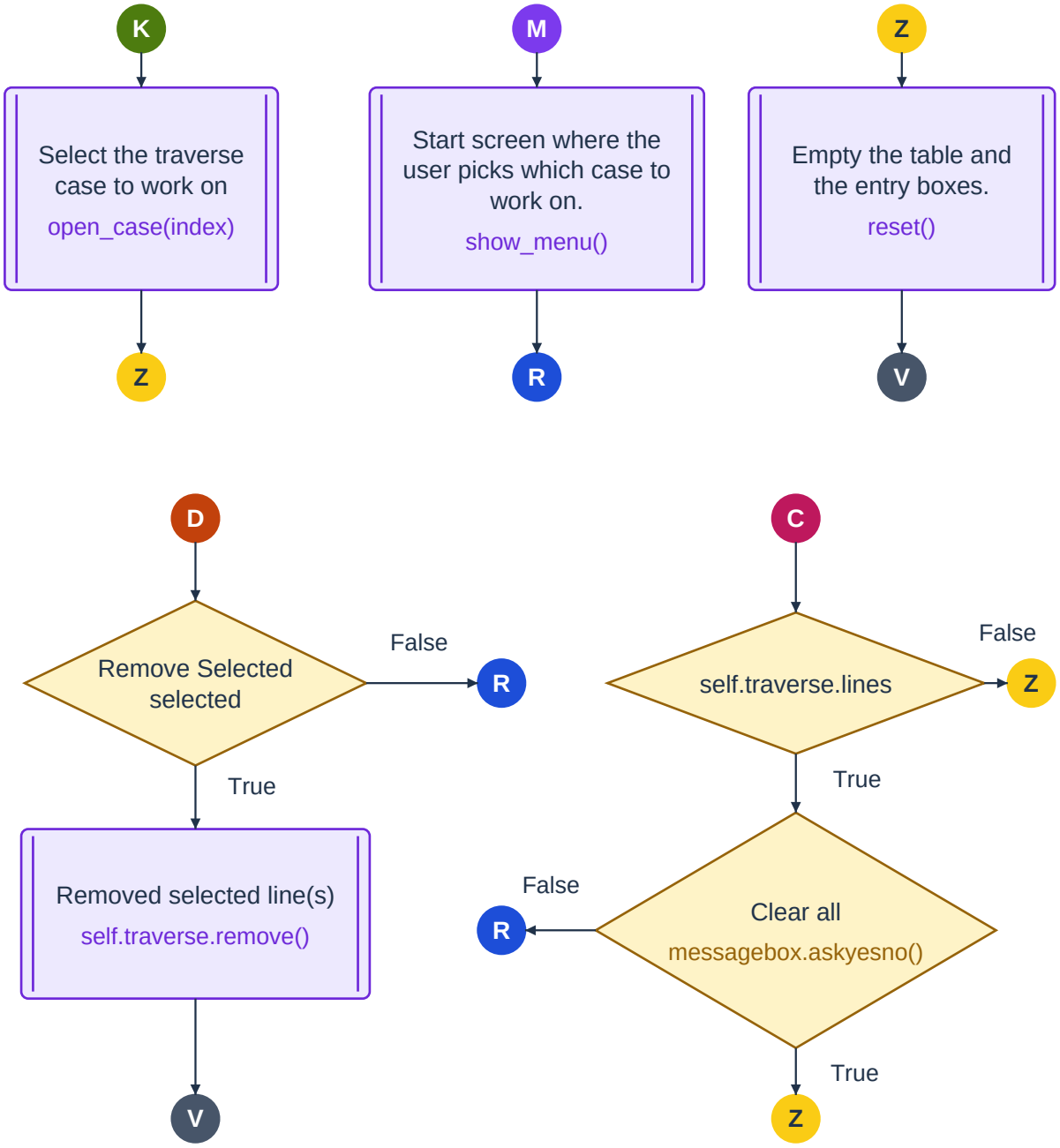
1. Add a traverse line. Enter Line, Bearing and Length (m). Use Unknown for omitted values.

2. Input fields. all(texts) rejects empty fields: Please fill in all input fields.

3. Build a Line from the entry boxes. Line.from_text() and Bearing.parse() validate the entries; ValueError shows Invalid input.

4. Add Line. self.traverse.add(line) saves the line; clear the entry boxes and continue at V.

3 MenuScreen / SolverScreen



NUMBERED STEPS

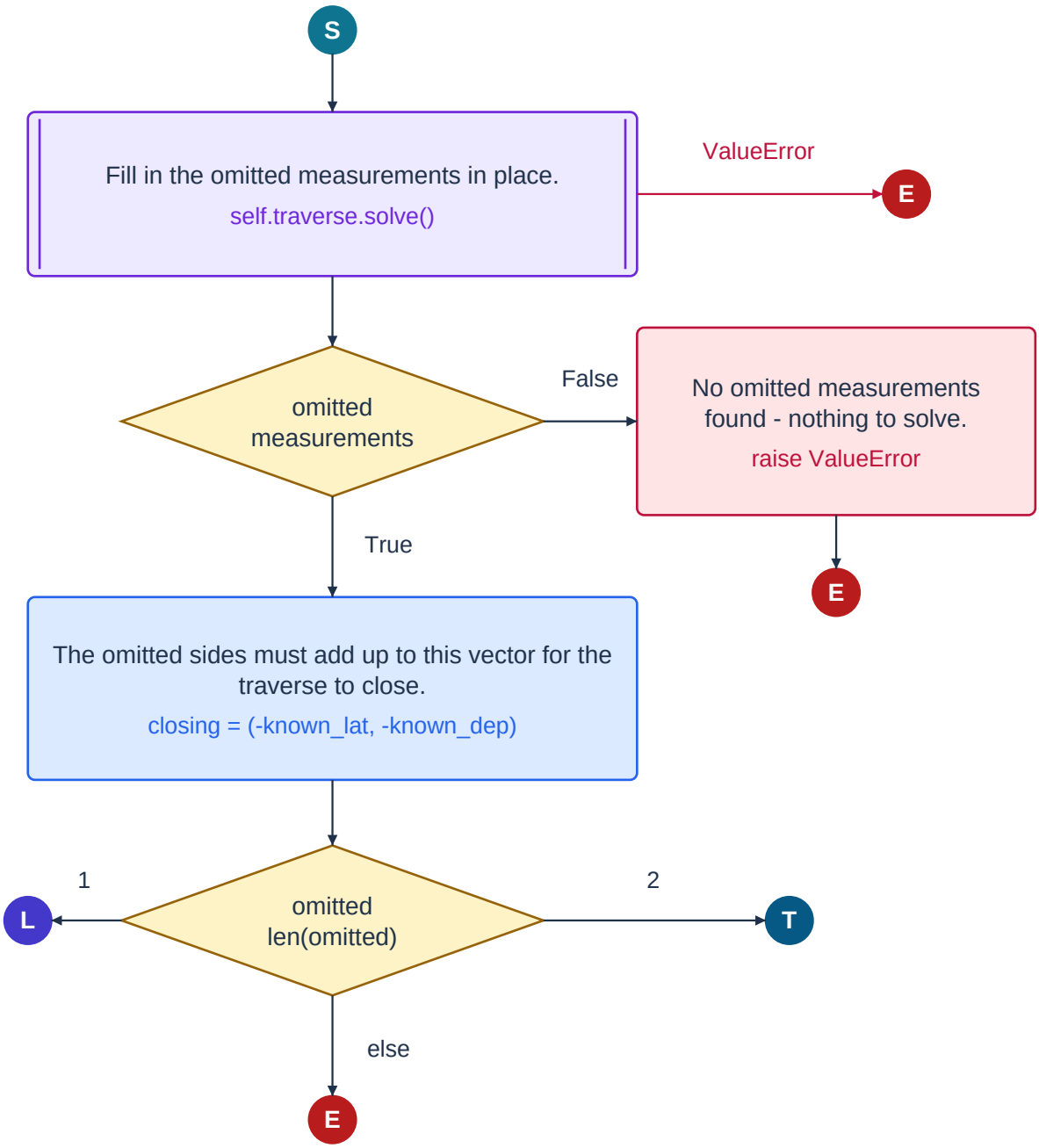
1. Select the traverse case to work on. open_case(index) switches to SolverScreen, sets the case and calls reset().

2. Return / Reset. show_menu() returns to the menu. reset(): Empty the table and the entry boxes.

3. Remove Selected. With selected rows, remove them and refresh at V. Without selected rows, return at R.

4. Clear All. An empty table resets directly. Otherwise, confirm Clear all: True resets at Z; False returns at R.

4 Solve Omitted Measurements



NUMBERED STEPS

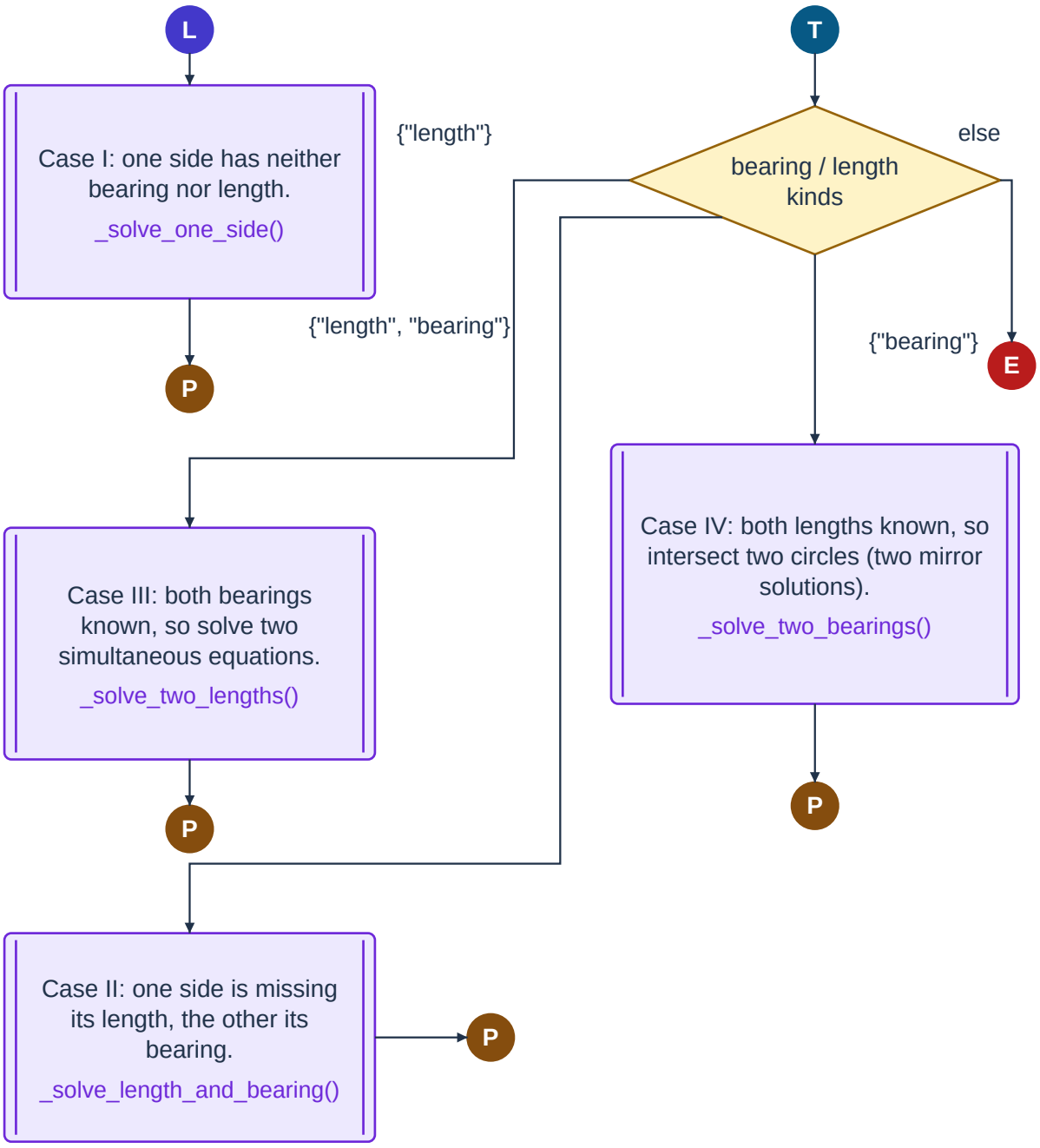
1. Fill in the omitted measurements in place. Call self.traverse.solve(); SolverScreen.solve() catches ValueError through E.

2. omitted. No omitted measurements found - nothing to solve. Otherwise, continue with the known lines.

3. closing. Use line.components() for complete lines; closing = (-known_lat, -known_dep).

4. len(omitted). One omitted side follows L; two follow T. Any other count follows E. The menu case does not choose the solver.

5 Omitted measurements



NUMBERED STEPS

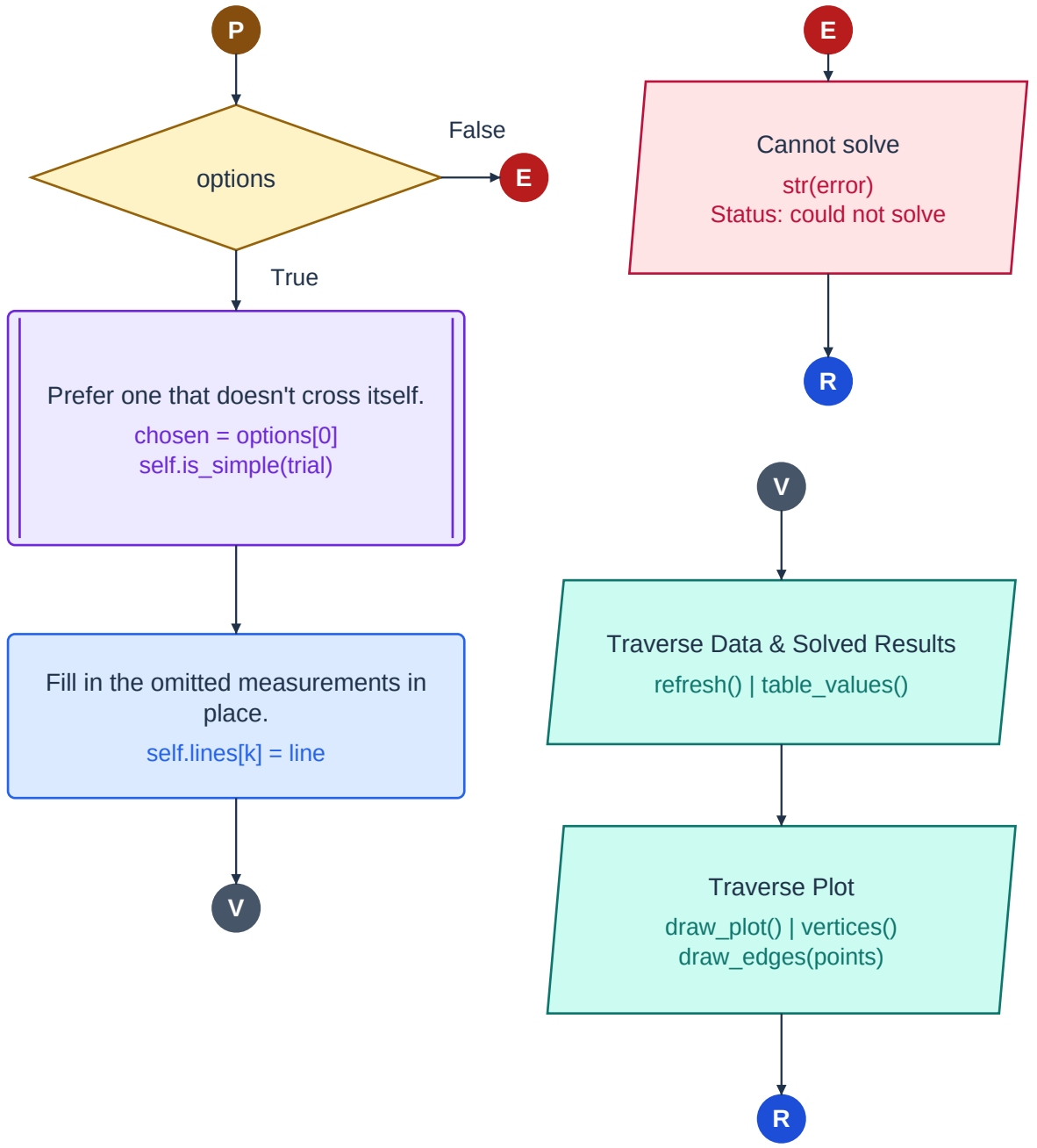
1. Case I. One side has neither bearing nor length: _solve_one_side(). Other one-side combinations raise ValueError.

2. Case III. Both bearings known, so solve two simultaneous equations: _solve_two_lengths().

3. Case IV. Both lengths known, so intersect two circles (two mirror solutions): _solve_two_bearings().

4. Case II. One side is missing its length, the other its bearing: _solve_length_and_bearing(). Unsupported kinds follow E.

6 Traverse Data & Solved Results



NUMBERED STEPS

1. options. No valid solution - check the input data. Empty options raise ValueError and follow E.

2. chosen. Prefer one that doesn't cross itself. Start with options[0]; keep it when no is_simple(trial) option is found.

3. Fill in the omitted measurements in place. Save chosen lines in self.lines[k], then refresh at V.

4. Traverse Data & Solved Results. refresh() updates the table and Traverse Plot. Cannot solve shows str(error); R resumes the event loop.